STAT 559 – Note 02
Outer Measures, Extension, and Completion

Edited transcription of handwritten lecture notes

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

Source page(s): 1–2

1 Outer measures and their measurable sets

The construction has three stages: extend to all subsets as an outer measure, restrict to the sets measurable for that outer measure, then restrict further to the desired generated sigma-algebra. An *outer measure* is $\phi : 2^\Omega \to [0, \infty]$ such that

$$\phi(\varnothing) = 0, \tag{1}$$

$$A \subseteq B \Longrightarrow \phi(A) \leq \phi(B), \tag{2}$$

$$\phi\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \phi(A_n). \tag{3}$$

Supplement: The source's question: is the power set a sigma-algebra?

Yes, 2^Ω is a sigma-algebra. An outer measure need not be a measure on it because countable *additivity* on disjoint sets is not assumed. Countable subadditivity alone does not give the volume contradiction discussed with Banach–Tarski. Outer measure is defined on all subsets; the power set is not assumed to be a proper subset of anything.

A set $A \subseteq \Omega$ is *Carathéodory-measurable for* ϕ if for every $E \subseteq \Omega$,

$$\phi(E) = \phi(E \cap A) + \phi(E \cap A^c). \tag{4}$$

Let $\mathcal{M}_\phi$ be the class of these sets. The central theorem is that $\mathcal{M}_\phi$ is a sigma-algebra and $\phi|_{\mathcal{M}_\phi}$ is a measure (reference Theorem 1.4.3).

Source page(s): 3–4

2 Proof: algebra and finite additivity

The empty set is measurable because $\phi(E) = 0 + \phi(E)$. The defining equality is symmetric in A, A^c, proving complement closure. For $A, B \in \mathcal{M}_\phi$, put $D = A \cup B$. Then

$$\phi(E \cap D) + \phi(E \cap D^c) \leq \phi(E \cap A) + \phi(E \cap A^c \cap B) + \phi(E \cap A^c \cap B^c) \tag{5}$$
$$= \phi(E \cap A) + \phi(E \cap A^c) = \phi(E). \tag{6}$$

The opposite inequality is outer subadditivity. Thus $D \in \mathcal{M}_\phi$ and $\mathcal{M}_\phi$ is an algebra. For disjoint $A_1, \ldots, A_n \in \mathcal{M}_\phi$ and any $E \subseteq \Omega$,

$$\phi\left(E \cap \bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \phi(E \cap A_i). \tag{7}$$

Source page(s): 5–6

Write $C_n = \bigcup_{i=1}^{n} A_i$. Splitting $E \cap C_n$ by measurable A_n gives

$$\phi(E \cap C_n) = \phi(E \cap A_n) + \phi(E \cap C_{n-1}), \tag{8}$$

which proves the identity by induction. If measurable $A_n \uparrow A$, the measurability of A is not yet known. Set $D_1 = A_1$ and $D_n = A_n \setminus A_{n-1}$ for $n \geq 2$. They are disjoint members of the algebra, and

$$\phi(E \cap A_n) = \sum_{i=1}^{n} \phi(E \cap D_i), \tag{9}$$

$$\lim_n \phi(E \cap A_n) = \sum_{i=1}^{\infty} \phi(E \cap D_i) \geq \phi(E \cap A). \tag{10}$$

Supplement: Passing from finite to infinite sums

The source asks how the limit is justified. For nonnegative summands the infinite series is, by definition, the increasing limit of its partial sums. The last inequality is countable subadditivity, not an application of measure continuity before the theorem has been proved.

Source page(s): 6–8

3 Proof: sigma-algebra and countable additivity

For arbitrary $A_i \in \mathcal{M}_\phi$, put $A = \bigcup_i A_i$ and $C_n = \bigcup_{i \leq n} A_i$. Since C_n is measurable and $A^c \subseteq C_n^c$,

$$\phi(E) = \phi(E \cap C_n) + \phi(E \cap C_n^c) \geq \phi(E \cap C_n) + \phi(E \cap A^c). \tag{11}$$

Taking the increasing limit and using the previous lemma gives

$$\phi(E) \geq \phi(E \cap A) + \phi(E \cap A^c). \tag{12}$$

Subadditivity gives equality. Thus $A \in \mathcal{M}_\phi$. For disjoint measurable A_i, monotonicity and finite additivity give

$$\phi\left(\bigcup_i A_i\right) \geq \sum_{i=1}^{n} \phi(A_i) \quad \text{for every } n. \tag{13}$$

Let $n \to \infty$ and combine with subadditivity to obtain countable additivity. The source also recalls the separate uniqueness step, proved by the pi–lambda theorem under a common sigma-finite cover.

Source page(s): 9–10

4 Premeasures and extension

A *premeasure* on an algebra $\mathcal{A}$ is a function $\mu_0 : \mathcal{A} \to [0, \infty]$ with $\mu_0(\varnothing) = 0$ and

$$\mu_0\left(\bigcup_i A_i\right) = \sum_i \mu_0(A_i) \tag{14}$$

whenever $A_i \in \mathcal{A}$ are disjoint and their union also belongs to $\mathcal{A}$. **Carathéodory extension theorem** (reference Theorem 1.5.1). A premeasure on an algebra extends to a measure on $\sigma(\mathcal{A})$. If $\Omega = \bigcup_n E_n$ for $E_n \in \mathcal{A}$ with $\mu_0(E_n) < \infty$, the extension is unique. Define, for arbitrary $S \subseteq \Omega$,

$$\mu^*(S) = \inf\left\{\sum_{i=1}^{\infty} \mu_0(C_i) : C_i \in \mathcal{A},\ S \subseteq \bigcup_{i=1}^{\infty} C_i\right\}. \tag{15}$$

The collection of covers is nonempty, since $\Omega, \varnothing, \ldots$ is available; the infimum can equal infinity.

Source page(s): 11–12

5 The covering functional is an outer measure

The empty cover by empty sets proves $\mu^*(\varnothing) = 0$. Every cover of T covers $S \subseteq T$, so $\mu^*(S) \leq \mu^*(T)$. To prove subadditivity, take arbitrary S_i. If $\sum_i \mu^*(S_i) = \infty$, the inequality is immediate. Otherwise, for $\varepsilon > 0$ choose covers $S_i \subseteq \bigcup_j C_{ij}$ with

$$\sum_j \mu_0(C_{ij}) < \mu^*(S_i) + 2^{-i}\varepsilon. \tag{16}$$

Their union is a countable cover of $\bigcup_i S_i$, hence

$$\mu^*\left(\bigcup_i S_i\right) \leq \sum_{i,j} \mu_0(C_{ij}) \tag{17}$$

$$\leq \sum_i \mu^*(S_i) + \varepsilon \sum_{i=1}^{\infty} 2^{-i} = \sum_i \mu^*(S_i) + \varepsilon. \tag{18}$$

Let $\varepsilon \downarrow 0$.

Original TODO: Check the definition of the infimum

The source marks the near-minimizing-cover step. For a finite infimum, some admissible cover has cost below the infimum plus any positive tolerance; otherwise that larger number would still be a lower bound. The choice $2^{-i}\varepsilon$ makes the total error at most ε. Nonnegative double sums may be enumerated in any order. The separate infinite case above avoids treating infinity as a finite approximation target. See reference Lemma 1.5.2.

Source page(s): 11, 13

Supplement: The homework step: agreement on the algebra

For $A \in \mathcal{A}$ the cover $A, \varnothing, \ldots$ gives $\mu^*(A) \le \mu_0(A)$. For any cover $A \subseteq \bigcup_i C_i$, disjointify $A \cap C_i$ within A. The resulting disjoint algebra sets D_i have union A, so

$$\mu_0(A) = \sum_i \mu_0(D_i) \le \sum_i \mu_0(C_i). \tag{19}$$

Taking the infimum gives $\mu_0(A) \le \mu^*(A)$. Thus equality holds (reference Lemma 1.5.3).

Source page(s): 13–14

6 Every algebra set is outer-measurable

Fix $A \in \mathcal{A}$, any $E \subseteq \Omega$, and an algebra cover $E \subseteq \bigcup_i C_i$. The sets $C_i \cap A$ cover $E \cap A$, and $C_i \cap A^c$ cover $E \cap A^c$. Finite additivity on $\mathcal{A}$ gives

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \le \sum_i [\mu_0(C_i \cap A) + \mu_0(C_i \cap A^c)] \tag{20}$$

$$= \sum_i \mu_0(C_i). \tag{21}$$

Take the infimum over covers. The reverse inequality is subadditivity, so $A \in \mathcal{M}_{\mu^*}$. Consequently $\sigma(\mathcal{A}) \subseteq \mathcal{M}_{\mu^*}$, and restriction gives the required measure extension.

Correction: The infimum is a lower envelope

The source's page 14 contains an intermediate inequality suggesting a cover cost is at most $\mu^*(E)$. Every cover cost is instead at least the infimum. The argument works because the same left side is bounded by *every* cover cost; taking their infimum gives the desired bound.

Source page(s): 15–16

7 Lebesgue measure and completion

Let $\mathcal{A}$ be the algebra of finite disjoint unions of half-open intervals $(a, b] \cap \mathbb{R}$, allowing extended endpoints with the natural ray conventions. Assign each bounded interval its length $b - a$ and each unbounded nonempty interval infinite length; sum over disjoint components. The source assigns as homework proving that $\mathcal{A}$ is an algebra and this functional is a premeasure. Reference Section 1.6

supplies this construction. Since $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$ and $\mathbb{R} = \bigcup_n (-n, n]$, the Borel extension is unique. Its completed version is the usual complete Lebesgue measure. The construction extends to $\mathbb{R}^n$; product measures are discussed later. A measure space is *complete* if every subset of every measurable null set is measurable.

Original TODO: A counterexample to automatic completeness

For a small explicit example, let $\Omega = \{a, b, c\}$ and $\mathcal{F} = \{\varnothing, \{a, b\}, \{c\}, \Omega\}$. Put mass zero on $\{a, b\}$ and mass one on $\{c\}$. Then $\{a\}$ is a subset of a measurable null set but is not in $\mathcal{F}$.

Supplement: The source's Borel-measurability question

Borel Lebesgue measure on $\mathcal{B}(\mathbb{R})$ is not complete. The Cantor set is Borel, null, and has cardinality $|\mathbb{R}|$. Its power set has strictly greater cardinality than the class of Borel sets, so some subsets of it are not Borel. They are measurable in the completion. This cardinality argument explains the example; the completion theorem itself is reference Theorem 1.7.2.

Source page(s): 17

8 Completing a measure space

For any measure space $(\Omega, \mathcal{F}, \mu)$, define

$$\overline{\mathcal{F}} = \{A \cup N : A \in \mathcal{F},\ N \subseteq Z \text{ for some } Z \in \mathcal{F},\ \mu(Z) = 0\}, \tag{22}$$

$$\overline{\mu}(A \cup N) = \mu(A). \tag{23}$$

Supplement: The completion homework

The formula is independent of the representation: two measurable parts differ only inside a measurable null set, so their measures agree. Complements and countable unions remain in this class by absorbing the exceptional subsets into countable unions of measurable null sets. Countable additivity follows by removing those null exceptional sets and using additivity on $\mathcal{F}$. Every subset of an $\overline{\mu}$-null measurable set is again of the indicated form, so this measure space is complete. Alternatively apply the covering construction to $\mathcal{F}$ itself and restrict to its Carathéodory-measurable sets, as proposed in the source. Every outer-null set is Carathéodory-measurable, which proves completeness of that resulting space (reference Section 1.7).